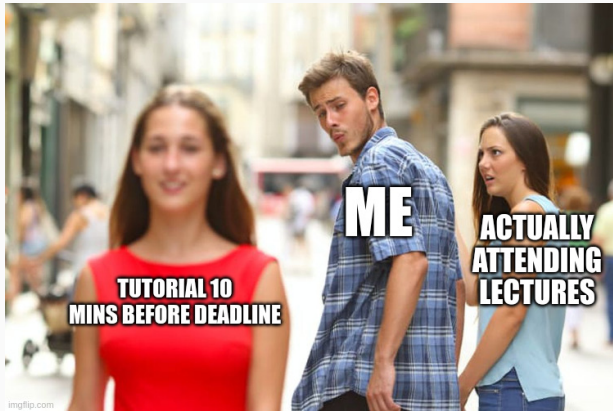


Discrete Structures Tutorial 2

Set Theory - Group 11

TA: Meet Parekh

Before We Start...



What We'll Cover

- Quick recap: Sets, Functions & Bijections

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- Quick recap: Sets, Functions & Bijections
- Comparing infinite sets (Hilbert's Hotel!)
- Countability + Cantor's Diagonalization
- All 5 Tutorial-2 problems, solved (DW, I'll share a few more with you too)

Sets: Ultra-Quick Recap

- Set = unordered collection. Example: $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$.
- $A \subseteq B$: every element of A is in B . (Here $A \not\subseteq B$, since $1 \notin B$.)
- $A \cup B = \{1, 2, 3, 4\}$, $A \cap B = \{2, 3\}$.

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- $A \cup B = \{1, 2, 3, 4\}$, $A \cap B = \{2, 3\}$.
- $\bar{A}$ = everything in the universe **not** in A .
- Power set $\mathcal{P}(A)$ = set of **all** subsets of A . If $|A| = n$, then $|\mathcal{P}(A)| = 2^n$. (Here $|\mathcal{P}(A)| = 2^3 = 8$.)

Functions & Bijections: Quick Recap

- $f : X \rightarrow Y$: every $a \in X$ hits **exactly one** $b \in Y$.

Functions & Bijections: Quick Recap

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- **Surjective**: every output is hit.
- **Bijective**: both at once.

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For finite X, Y :

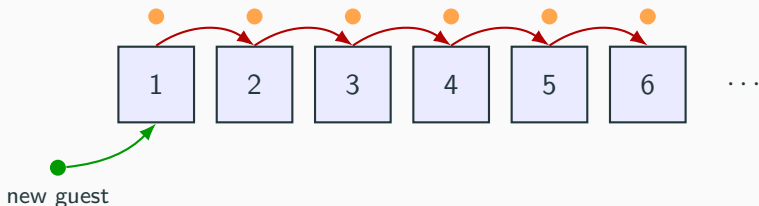
- injective $\Rightarrow |X| \leq |Y|$
- surjective $\Rightarrow |X| \geq |Y|$
- bijective $\Rightarrow |X| = |Y|$

Hilbert's Hotel: Act 1



- Infinite hotel. **Every room is full.**

Hilbert's Hotel: Act 1



- **One new guest arrives.** Room $n \rightarrow$ Room $n + 1$, for everyone.
- Room 1 is now free. *No one was kicked out.*

Theorem 1: The Shift, Formally

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A is infinite $\iff$ there is a bijection $A \rightarrow A \cup \{b\}$ (for $b \notin A$).

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($\Leftarrow$, **contrapositive**): If A is finite, no such bijection exists:
pigeonhole ($A \cup \{b\}$ has one more element).

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($\Leftarrow$, **contrapositive**): If A is finite, no such bijection exists: pigeonhole.

($\Rightarrow$): A infinite. Pick $a_0 \in A$, set $f(a_0) = b$. Pick $a_1 \in A \setminus \{a_0\}$ (nonempty since A infinite), set $f(a_1) = a_0$. In general $f(a_i) = a_{i-1}$. For every other $a \in A$: $f(a) = a$. This chain never runs out precisely *because* A is infinite, that's the only place infiniteness gets used.

Theorem 1: The Shift, Formally

Theorem

A is infinite $\iff$ there is a bijection $A \rightarrow A \cup \{b\}$ (for $b \notin A$).

$(\Leftarrow)$: finite sets can't absorb an extra element, pigeonhole. $(\Rightarrow)$: infinite sets always can, via the explicit shift above.

So: both directions together give the iff. An infinite set can always “make room” for one more element without changing size, a finite one never can. *This is the formal engine behind the Act 1 room-shift you just saw.*

Countable & Countably Infinite

- C is **countably infinite** if there's a bijection $C \rightarrow \mathbb{N}$.
- C is **countable** if it's finite OR countably infinite.
- Countably infinite = the “smallest” kind of infinite.

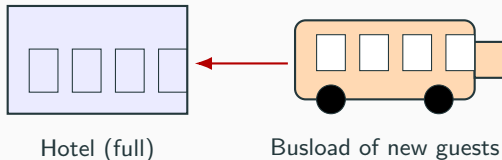
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Examples:

- Countably infinite: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, even numbers $2\mathbb{N}$ (bijection $n \mapsto 2n$).
- Countable (but finite): $\{1, 2, 3\}$, any finite set.

Hilbert's Hotel: Act 2

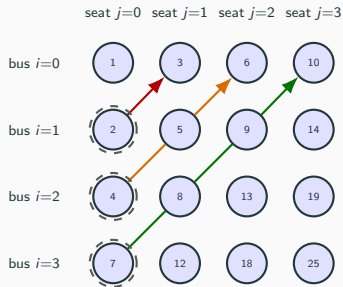


- One busload of infinitely many new guests arrives.
- Guest currently in **Room** n moves to **Room** $2n$.
- The n -th new arrival (1st, 2nd, 3rd, ... off the bus) goes to **Room** $2n - 1$.
- Odds for new folks, evens for old, everyone fits again.

Not a Bad Trade



Hilbert's Hotel: Act 3



- Now **infinitely many buses** arrive, **each** with infinitely many guests.
- Bus i , seat $j = \text{pair } (i, j)$. Walk the grid **diagonally**, numbering rooms as you go.

Quick Note: Act 3 Only Needs an Injection

- The Hotel's actual job here: give every guest, across every bus, a room of their own, nobody shares. That's an **injection** $(i,j) \mapsto \text{room}$, nothing more.
- It does **not** require every room number to get used, that would be asking for surjectivity too.
- Theorem 2 (next) proves something stronger: a full **bijection**. We don't strictly need that just to solve the hotel, but the numbering we're about to build gives it to us for free.

Theorem 2: Infinitely Many Buses, Formally

Theorem

$\mathbb{N} \times \mathbb{N}$ is countably infinite.

(So: the product of two countably infinite sets is countable.)

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The idea: don't number pair by pair, number **diagonal by diagonal**.

- Group pairs (i, j) by $d = i + j$. Diagonal d has exactly $d+1$ pairs, running from $(d, 0)$ (**left** end) to $(0, d)$ (**top** end).
- Sweep **left to top** within a diagonal, then move to the next diagonal. Every pair gets visited exactly once.

Theorem 2: Infinitely Many Buses, Formally

Theorem

$\mathbb{N} \times \mathbb{N}$ is countably infinite.

Term 1, how many rooms came before? Let $T_d = \sum_{k=1}^d k$.
Diagonals $0, 1, \dots, d-1$ together hold exactly T_d pairs (diagonal $k-1$ contributes k pairs, for $k = 1, \dots, d$).

So if (i, j) sits on diagonal $d = i + j$, then T_{i+j} pairs were already numbered before we even reach this diagonal.

Theorem 2: Infinitely Many Buses, Formally

Theorem

$\mathbb{N} \times \mathbb{N}$ is countably infinite.

Term 2, where am I within my own diagonal? Counting from the left end ($j = 0$), pair (i, j) sits in position $j + 1$.

Put the two terms together:

$$f(i, j) = \underbrace{\sum_{k=1}^{i+j} k}_{\text{rooms used by earlier diagonals}} + \underbrace{(j+1)}_{\text{position within this diagonal}}$$

Theorem 2: Infinitely Many Buses, Formally

Why is this a **bijection** $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$?

- **Injective:** different diagonals never share a value, diagonal $d+1$'s smallest output is exactly 1 more than diagonal d 's largest. Within one diagonal, different j give different positions.
- **Surjective:** every $n \in \mathbb{N}$ falls into exactly one diagonal d (the one with $T_d < n \leq T_{d+1}$), which fixes $j = n - T_d - 1$ and $i = d - j$.

Injective + surjective $\Rightarrow$ **bijection**.

Problem 1 (Finite Set)

Let $A = \{f : \{1, 2, \dots, n\} \rightarrow \{0, 1\} \mid f \text{ is injective}\}.$

- (a) Express $|A|$ in terms of n .
- (b) Prove that A is finite for any fixed $n \in \mathbb{N}$.
- (c) What happens to $|A|$ as n increases beyond 2?

Problem 1: Solution

$$A = \{f : \{1, \dots, n\} \rightarrow \{0, 1\} \mid f \text{ is injective}\}$$

(a) Express $|A|$ in terms of n .

- Each domain element needs a **distinct** image, but the codomain has only 2 elements.
- $n = 1$: 2 choices $\Rightarrow |A| = 2$. $n = 2$: 2 valid arrangements $\Rightarrow |A| = 2$.
- $n \geq 3$: impossible $\Rightarrow |A| = 0$.

Problem 1: Solution

$$A = \{f : \{1, \dots, n\} \rightarrow \{0, 1\} \mid f \text{ is injective}\}$$

(a) $|A| = 2$ for $n \in \{1, 2\}$; $|A| = 0$ for $n \geq 3$.

(b) Prove A is finite.

$A \subseteq \{\text{all functions } \{1, \dots, n\} \rightarrow \{0, 1\}\}$, and that full set has size 2^n , finite. A subset of a finite set is finite $\Rightarrow A$ finite.

(c) What happens as n grows past 2?

For $n \geq 3$: injective $f : X \rightarrow Y$ (finite) needs $|X| \leq |Y|$. Here $|X| = n \geq 3 > 2 = |Y|$, contradiction. So $A = \emptyset$.

Problem 2 (Finite Set)

Consider the set $B = \{S \subseteq \mathbb{N} \mid S \text{ is finite and } \sum_{s \in S} s \leq 100\}$.

- (a) Prove that B is finite.
- (b) Determine whether the set $C = \{S \subseteq \mathbb{N} \mid S \text{ is finite and } |S| \leq 5\}$ is finite or infinite.
- (c) Compare the cardinalities of B and C .

Problem 2: Solution

$$B = \{S \subseteq \mathbb{N} \mid S \text{ finite}, \sum_{s \in S} s \leq 100\}$$

$$C = \{S \subseteq \mathbb{N} \mid S \text{ finite}, |S| \leq 5\}$$

(a) Prove B finite. Every $s \in S$ satisfies $s \leq 100$ (sum of positive terms $\geq$ any one term). So $S \subseteq \{1, \dots, 100\}$ always $\Rightarrow B \subseteq \mathcal{P}(\{1, \dots, 100\})$, size 2^{100} . Finite!

(b) C finite or infinite? Infinite: $n \mapsto \{n\}$ is an injection $\mathbb{N} \hookrightarrow C$ (every singleton has size $1 \leq 5$).

(c) Compare $|B|$ and $|C|$. B finite, C infinite $\Rightarrow |B| < |C|$.

Problem 3 (Infinite Set)

Let $D = \{(a, b) \in \mathbb{Q} \times \mathbb{Q} \mid a^2 + b^2 = 2\}$.

- (a) Prove that D is infinite.
- (b) Is D countably infinite or uncountably infinite? Justify your answer.
- (c) Construct an explicit bijection between D and a proper subset of itself.

Problem 3: Solution

$$D = \{(a, b) \in \mathbb{Q} \times \mathbb{Q} \mid a^2 + b^2 = 2\}$$

(a) Prove D is infinite.

- Spot one rational solution: $(1, 1)$. Check: $1^2 + 1^2 = 2$. ✓
- **Idea:** draw a line of rational slope t through $(1, 1)$; it hits the circle at exactly one *other* point. Different t give different points.

Problem 3(a): Finding the Family

Line through $(1, 1)$ with rational slope t : $b - 1 = t(a - 1)$, i.e.
 $b = t(a - 1) + 1$.

Substitute into $a^2 + b^2 = 2$:

$$a^2 + (t(a - 1) + 1)^2 = 2 \implies a^2 + t^2(a - 1)^2 + 2t(a - 1) - 1 = 0$$

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Since $a^2 - 1 = (a - 1)(a + 1)$, factor out $(a - 1)$:

$$(a - 1) \left[(a + 1) + t^2(a - 1) + 2t \right] = 0$$

So $a = 1$ (the point we already knew), **or** the bracket is zero.

Problem 3(a): Finding the Family

From the bracket = 0: $a(1 + t^2) = t^2 - 2t - 1 \Rightarrow a = \frac{t^2 - 2t - 1}{1 + t^2}$

Plug back into $b = t(a - 1) + 1$:

$$b = \frac{-t^2 - 2t + 1}{1 + t^2}$$

$$\boxed{a(t) = \frac{t^2 - 2t - 1}{1 + t^2}, \quad b(t) = \frac{-t^2 - 2t + 1}{1 + t^2}} \quad t \in \mathbb{Q}$$

Different $t \Rightarrow$ different lines through $(1, 1) \Rightarrow$ different second points. $\mathbb{Q}$ is infinite $\Rightarrow$ **D is infinite.**

Problem 3(b), (c)

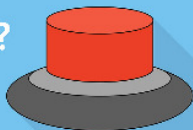
(b) Countably or uncountably infinite?

- $D \subseteq \mathbb{Q} \times \mathbb{Q}$, and $\mathbb{Q} \times \mathbb{Q}$ is countable: **that's Theorem 2 from the hotel!**
- Subset of countable is countable. D infinite + countable = **countably infinite.**

(c) Bijection to a proper subset of D .

- D countably infinite $\Rightarrow$ enumerate $D = \{d_1, d_2, d_3, \dots\}$.
- Define $h(d_i) = d_{i+1}$. This is a bijection $D \rightarrow D \setminus \{d_1\}$.
- **This is the Act-1 room shift, applied to D itself!**

WILL YOU PRESS
THE BUTTON?



Countably
Infinite

but

Uncountable



Problem 4 (Infinite Set)

Let $E = \{n \in \mathbb{N} \mid n \equiv 1 \pmod{3}\} \cup \{n \in \mathbb{N} \mid n \equiv 2 \pmod{5}\}$.

- (a) Write out the first 10 elements of E .
- (b) Prove that E is infinite.
- (c) Is E countably infinite? Justify your answer.

Problem 4: Solution

$$E = \{n \in \mathbb{N} \mid n \equiv 1 \pmod{3}\} \cup \{n \in \mathbb{N} \mid n \equiv 2 \pmod{5}\}$$

(a) First 10 elements: 1, 2, 4, 7, 10, 12, 13, 16, 17, 19

(7 and 22 appear in both lists, don't double count!)

(b) Prove E infinite. $E \supseteq \{1, 4, 7, 10, \dots\}$ (via $n \mapsto 3n - 2$), already infinite $\Rightarrow$ superset E is infinite too.

(c) Is E countably infinite? $E \subseteq \mathbb{N}$, $\mathbb{N}$ countable $\Rightarrow E$ countable (subset-of-countable). Infinite + countable = **countably infinite**.

Plot Twist Incoming

**COUNTING
ON FINGERS**

**COUNTING
INFINITE SETS**

BIJECTIONS

**SOME
INFINITES ARE
BIGGER THAN OTHERS**



Problem 5 (Cantor's Diagonalization)

Consider the set $T = \{f : \mathbb{N} \rightarrow \{0, 1\}\}$ of all functions from natural numbers to $\{0, 1\}$.

- (a) Assume T is countably infinite and list the functions as $f_1, f_2, f_3, \dots$
- (b) Define a new function $g : \mathbb{N} \rightarrow \{0, 1\}$ by $g(n) = 1 - f_n(n)$.
- (c) Show that g cannot be in the list $\{f_1, f_2, f_3, \dots\}$.
- (d) What does this contradiction tell us about the cardinality of T ?

Problem 5: Solution

$$T = \{f : \mathbb{N} \rightarrow \{0, 1\}\}$$

(a) Suppose T is countably infinite: list it as $f_1, f_2, f_3, \dots$

(b) Build a new function: $g(n) = 1 - f_n(n)$

(flip the n -th entry of the n -th function: walk the diagonal)

Problem 5: Solution

$$T = \{f : \mathbb{N} \rightarrow \{0, 1\}\}$$

(a) Suppose T countably infinite: list it as $f_1, f_2, f_3, \dots$

(b) $g(n) = 1 - f_n(n)$

(c) **Could $g = f_k$ for some k ?** Then $g(k) = f_k(k)$. But by definition $g(k) = 1 - f_k(k)$. So $f_k(k) = 1 - f_k(k) \Rightarrow f_k(k) = \frac{1}{2}$. **Impossible.**

(d) $g \in T$ but g isn't in the list, yet the list was supposed to contain *all* of T . **T is uncountable.**

*This time, no matter how you number the rooms, someone's
always left outside the hotel.*

Alright, We're Almost Done



Cramming
the night
before exam



Coming
to TA hours
with doubts

A Few Things Before You Go

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- Use AI tools if you want, but verify the logic. No doubt is too small.

TA: Meet Parekh

TA Hours: Saturday, 3:00 PM – 4:00 PM

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- **Drop your doubt on WhatsApp before TA hours:** saves both our time, we go in ready.
- Not at my spot during TA hours? Message me, I'll come find you.

Thank You!

- Sorry if we rushed at the end!
- Drop me a DM with feedback (too fast? too slow? more memes?).



30-second anonymous feedback: Helpful? Not helpful? Anything else?

<https://forms.gle/Dhq9pSdo64vuGbye8>